

Samsung SSD 950 PRO M.2

Data Sheet Rev.1.2 (June 2016)



Summary

- PCIe 3.0 x4 NVMe Express SSD for Client PCs
- M.2 (2280)
- Samsung V-NAND
- Samsung Magician Software for SSD management

V-NAND Technology and THE SAMSUNG SSD 950 PRO

Samsung's unique and innovative V-NAND flash memory architecture is a breakthrough in overcoming the density limitations, performance and endurance of today's conventional planar NAND architecture. V-NAND is fabricated by stacking layers vertically over one another rather than decreasing the cells dimensions and trying

to fit itself onto a fixed horizontal space resulting in higher density and better performance utilizing a smaller footprint.

This cutting-edge V-NAND-based NVMe SSD supports PCI Express® Gen 3 x4 lanes, providing a higher bandwidth and lower latency to process a massive amount of more data than previous-generation SATA SSDs. Presented in a compact M.2 form factor, the future-oriented 950 PRO is ideal for professionals who want cutting-edge performance from their high-end PCs and workstations.

Experience next-generation ultimate SSD today

The 950 PRO NVMe SSD is the ultimate flash storage solution, designed with tomorrow's technology today. Providing an enhanced bandwidth, it's ideal for intensive workloads, such as computer-aided design, data analysis and engineering simulations. It

outperforms SATA SSDs with 4.5X faster sequential read (2,500 MB/s) and 2.5X faster sequential write speeds (1,500 MB/s). Plus, the 950 PRO boasts random read/write speeds of up to whopping 300K/110K IOPS.

Maximize endurance reliability under heavy workloads

Maximize endurance under heavy workloads

The innovative V-NAND technology designed into the 950 PRO strengthens the endurance of the SSD through insulators that cause less stress and are more resistant to wear, prolonging the lifespan of the SSD. This state-of-the-art technology has been tested for endurance on client systems under heavy workloads over a 5-year warranty period. These tests ensure that the 950 PRO will maintain a high level of sustained performance for long periods of time under heavy workloads..

Save power while maintaining peak performance

If you're in the midst of performing a complex data analysis or engineering simulation on your laptop, the last thing you want to have happen is your battery runs out. Designed to consume 5.7 W at peak workloads, the 950 PRO secures battery life while maintaining superior performance levels. The Active State Power Management (ASPM) mechanism built into the 950 PRO, which is similar to the device sleep mode in a SATA interface, reduces power consumption. In standby mode, the 950 PRO consumes only 2.5 mW of power, half the power consumption of an SATA (compared to 850 PRO) in Device Sleep Mode.

Dynamic Thermal Throttling Protection

In most cases of data transfers, heavy workloads can induce heat and result in high temperatures. Once temperatures reach a threshold, memory performance may malfunction. As the leader in SSD memory solutions, we equipped the 950 PRO with Dynamic Thermal Throttling Protection technology to automatically monitor and maintain optimal operating temperature to reduce the risks of overheating.

Technical Specifications

Samsung SSD 950 PRO			
Usage Application	Client PCs		
Capacity	256GB, 512GB		
Dimensions (LxWxH)	Max 80.15 x Max 22.15 x Max.2.38 (mm)		
Interface	PCIe 3.0 x4 (up to 32Gb/s) NVMe 1.1		
Form Factor	M.2(2280) [†]		
Controller	Samsung UBX controller		
NAND Flash Memory	Samsung V-NAND		
DRAM Cache Memory	512MB LPDDR3		
Performance*		256GB	512GB
	Sequential Read:	Up to 2,200 MB/s	Up to 2,500 MB/s
	Sequential Write**:	Up to 900 MB/s	Up to 1,500 MB/s
	4KB Random Read (QD32 Thread 4)	Up to 270K IOPS	Up to 300K IOPS
	4KB Random Write(QD32 Thread 4)	Up to 85K IOPS	Up to 110K IOPS
	4KB Random Read(QD1 Thread 1):	Up to 11K IOPS	Up to 12K IOPS
	4KB Random Write(QD1 Thread 1):	Up to 43K IOPS	Up to 43K IOPS
Data Security	AES 256-bit for User Data Encryption ^{††}		
Weight	Max. 10g (512GB)		
Reliability	MTBF : 1.5 million hours		
Power Consumption**	Active Read/Write (Average): Max. 5.6W(512GB) / Max.5.7 W(512GB) Idle : 70mW DEVSLP(L1.2 mode) : 2.5mW		
Supporting features	TRIM(Required OS support), Garbage Collection, S.M.A.R.T		
Temperature	Operating Temp : 0°C to 70°C (Measured by SMART Temperature. Proper airflow recommended)		
Humidity	5% to 95%, non-condensing		
Vibration	Non-Operating: 20~2000Hz, 20G		
Shock	Non-Operating: 1500G , duration 0.5m sec, 3 axis		
Warranty	5 years limited		
TBW	256GB : 200Terabytes-Written(TBW), 512GB : 400TBW		

* Sequential performance measurements based on Crystal Disk Mark 5.0.2 and Random performance measurements based on Iometer 1.1.0. Performance may vary based on SSD's firmware version, system hardware & configuration. Test system configuration : Intel Core i5-6600K @ 3.5GHz, DDR3 1600MHz 8GB, OS – Windows10 x64, Mainboard: MSI (Skylake), Model : Z170 KRAIT Gaming

**Power consumption measured with Iometer 1.1.0 with Intel i7-4770K(Haswell, 3.5GHz), DDR3 8GB, ASRock Z87 Extreme9/ac, APST on, OS- Windows7 Ultimate x64 SP1

†M.2 is a specification of form factor for ultra-thin PCs, The M.2 standard allows widths 12, 16, 22 and 30mm and lengths of 16, 26, 30, 38, 42, 69, 80 and 110 mm, Commercially M.2 is popular with width 22mm and lengths 30, 42, 60, 80 and 110mm. Samsung provides the most popular formfactor with 22mm X 80mm model (i.e., 2280) to consider user convenience.

†† The plan to provide a firmware update to enable TCG/OPAL and IEEE1667 has been put on hold due to the currently very restricted availability of commercial security software.

Product Lineup

Density	Model Name	Box Contents	Model Code
256 GB*	MZ-VKV256	Samsung SSD 950 PRO 256GB Warranty statement	MZ-V5P256BW
512 GB*	MZ-VKV512	Samsung SSD 950 PRO 512GB Warranty statement	MZ-V5P512BW

* GB: 1GB = 1,000,000,000 bytes. A certain portion of capacity may be used for system file and maintenance use, thus the actual capacity may differ that indicated on the product label.

For more information, please visit

www.samsung.com/ssd and www.samsungssd.com

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Samsung V-NAND SSD 980

2021 Data Sheet

Revision 1.1



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Revision History

Revision Number	Description	Revision Date
1.0	Initial Release	February 2021
1.1	Number of thread under random QD 32 performance revised from 16 to 4	June 2021

TECHNICAL SPECIFICATIONS

Samsung SSD 980					
Usage Application	Client PCs				
Interface	PCIe Gen 3.0 x4, NVMe 1.4				
Hardware Information	Capacity ¹⁾		250GB	500GB	1TB
	Controller		Samsung in-house Controller		
	NAND Flash Memory		Samsung V-NAND 3bit MLC		
	DRAM Cache Memory		-		
	Dimension		Max 80.15 x Max 22.15 x Max 2.38 (mm)		
	Form Factor		M.2 (2280)		
Performance (Up to.) ^{2) 3)}	Sequential Read		2,900 MB/s	3,100 MB/s	3,500 MB/s
	Sequential Write		1,300 MB/s	2,600 MB/s	3,000 MB/s
	QD 1 Thread 1	Ran. Read	17K IOPS	17K IOPS	17K IOPS
		Ran. Write	53K IOPS	54K IOPS	54K IOPS
	QD 32 Thread 4	Ran. Read	230K IOPS	400K IOPS	500K IOPS
		Ran. Write	320K IOPS	470K IOPS	480K IOPS
Power Consumption (Up to) ⁴⁾	Idle (ASPT on)		45mW		
	Active (Avg.)	Read	3.7 W	4.3 W	4.5 W
		Write	3.2 W	4.2 W	4.6 W
	L1.2 mode		5 mW		
Reliability	Temp.	Operating	0°C to 70°C (Measured by S.M.A.R.T. Temperature Proper airflow recommended)		
		Non-Operating	-40°C to 85°C		
	Humidity		5% to 95% non-condensing		
	Shock	Non-Operating	1,500G(Gravity), duration: 0.5ms, 3 axis		
	Vibration	Non-Operating	20~2,000Hz, 20G		
	MTBF		1.5 million hours		
	Warranty ⁸⁾	TBW ^{5) 6)}		150TB	300TB
Period		5 years limited ⁷⁾			
Supporting Features	TRIM (Required OS support), Garbage Collection, S.M.A.R.T				
Data Security	AES 256-bit Full Disk Encryption, TCG/Opal V2.0, Encrypted Drive (IEEE1667)				

1) 1GB = 1,000,000,000 bytes by IDEMA. A certain portion of capacity may be used for system file and maintenance use, thus the actual available capacity may differ from the labeled capacity.

2) Sequential and random performance measurements are based on IOMeter1.1.0. Performance may vary based on SSD's firmware version, system hardware & configuration. Test System: Intel(R) Core(TM) i7-6700K CPU @ 4.00GHz, DDR4 2133MHz 8GBx2 OS-Windows 10 Pro 64bit Asrock Z-170 Extreme7+

3) Sequential and random write performance was measured with Intelligent TurboWrite technology being activated. Intelligent TurboWrite operates only within a specific data transfer size. Performance may vary depending on SSD's firmware, system hardware & configuration and other factors. For detailed information, please contact your local service center.

4) Power consumption is measured with IOMeter1.1.0 version with AMD Ryzen 7 3700X 8 Core @3.6GHz, DDR4 8GBx2, OS-Windows 10 Pro 64bit, Chipset-GIGABYTE-X570-AORUS MASTER

5) All documented endurance test results are in compliance with JESD218 Standards. Please visit www.jedec.org for detailed information on JESD218 Standards.

6) TBW means Total Bytes Written.

7) Please refer to the detailed warranty statement here at <http://www.samsung.com/samsungssd>.

8) Warranty provides coverage for the stated time period or the TBW, whichever comes first.

PRODUCT LINEUP

Density	Model Name	Box Contents	Model Code
250GB*	MZ-V8V250	Samsung SSD 980 250GB Warranty Statement	MZ-V8V250BW MZ-V8V250B/AM
500GB*	MZ-V8V500	Samsung SSD 980 500GB Warranty Statement	MZ-V8V500BW MZ-V8V500B/AM
1TB (1,000GB*)	MZ-V8V1T0	Samsung SSD 980 1TB Warranty Statement	MZ-V8V1T0BW MZ-V8V1T0B/AM

* GB: 1GB = 1,000,000,000 bytes. The actual usable capacity may be less than the labeled capacity.

For more information, including but not limited to the warranty provided for this product, and to download the latest software & manuals, please visit www.samsung.com/ssd and www.samsungssd.com.

TEST CONFIGURATION

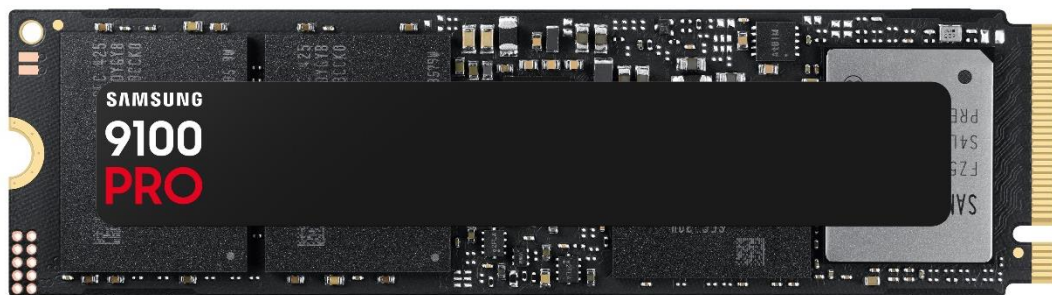
Below you will find a list of system configurations Samsung used to obtain the results reported in this Data Sheet. All performance data was measured with the SSD as a secondary drive

	Read/Write Performance	Power Consumption
Interface	PCIe Gen 3.0 x4	PCIe Gen 3.0 x4
OS	Windows 10 Pro 64bit	Windows 10 Pro 64bit
CPU	Intel(R) Core™ i7-6700K CPU@4.00GHz	AMD Ryzen 7 3700X 8 Core @3.6GHz
Memory	DDR4 2133MHz 8GBx2	DDR4 8GBx2
Chipset	Asrock Z-170 Extreme 7+	GIGABYTE-X570-AORUS MASTER
Test Program	IOmeter 1.1.0	IOmeter 1.1.0

Samsung V-NAND SSD 9100 PRO

2025 Data Sheet

Revision 2.0



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To download the latest software & manuals, please visit www.samsung.com/samsungssd

TECHNICAL SPECIFICATIONS

Samsung SSD 9100 PRO						
Usage Application	Client PCs, PlayStation®5					
Interface	PCIe® 5.0 x4, NVMe™ 2.0					
Hardware Information	Capacity ¹⁾		1TB	2TB	4TB	8TB
	Controller		Samsung in-house Controller			
	NAND Flash Memory		Samsung V-NAND TLC			
	Cache Memory		1GB LPDDR4X	2GB LPDDR4X	4GB LPDDR4X	8GB LPDDR4X
	Dimension ²⁾		1TB/2TB/4TB : Max 80.15 x Max 22.15 x Max 2.38 (mm) 8TB : Max 80.15 x Max 22.15 x Max 3.88 (mm)			
	Form Factor		M.2 (2280)			
Performance (Up to.) ^{3) 4) 5)}	Sequential Read		14,700 MB/s	14,700 MB/s	14,800 MB/s	14,800 MB/s
	Sequential Write		13,300 MB/s	13,400 MB/s	13,400 MB/s	13,400 MB/s
	QD 256 Thread 32	Ran. Read	1,850 KIOPS	1,850 KIOPS	2,200 KIOPS	2,200 KIOPS
		Ran. Write	2,600 KIOPS	2,600 KIOPS	2,600 KIOPS	2,600 KIOPS
Power Consumption	Active ⁶⁾⁷⁾ (Avg.)	Read	7.6 W	8.1 W	9.0 W	10.5 W
		Write	7.2 W	7.9 W	8.2 W	8.8 W
	Idle (Typical)	PS3(APST on)	4.0 mW	4.8 mW	6.5 mW	9.3 mW
		PS4 (L1.2)	3.3 mW	4.0 mW	5.7 mW	8.6 mW
Reliability	Temp.	Operating	0°C to 70°C (Measured by S.M.A.R.T. Temperature Proper airflow recommended)			
		Non-Operating	-40°C to 85°C			
	Humidity		5% to 95% non-condensing			
	Shock	Non-Operating	1,500G(Gravity), duration: 0.5ms, 3 axis			
	Vibration	Non-Operating	20~2,000Hz, 20G			
	MTBF		1.5 million hours			
Warranty ⁸⁾	TBW		600 TB	1,200 TB	2,400 TB	4,800 TB
	Period		5 years limited			
Supporting Features	TRIM (Required OS support), Garbage Collection, S.M.A.R.T					
Data Security	AES 256-bit Full Disk Encryption, TCG/Opal V2.0, Encrypted Drive (IEEE1667)					

1) 1GB = 1,000,000,000 bytes by IDEMA. A certain portion of capacity may be used for system file and maintenance use, thus the actual available capacity may differ from the labeled capacity.

2) The 8TB model complies PCI-SIG® D5 standard specification. Please check the specifications of the host device before installation.

3) 9100 PRO is backward compatible with PCIe 4.0 x4 and 3.0 x4.

4) Sequential and random performance measurements are based on FIO 3.33. Performance may vary based on SSD's firmware version, system hardware & configuration. Test System: AMD Ryzen 9 7950X 16-Core Processor CPU@4.50GHz, DDR5 4800MHz 16GBx2, OS-Ubuntu 22.04.2, Chipset-ASRock-X670E-Taichi.

5) Sequential and random write performance was measured with Intelligent TurboWrite technology being activated. Intelligent TurboWrite operates only within a specific data transfer size. For detailed information, please contact your local service center.

6) Active power consumption is measured on FIO 3.33 with AMD Ryzen 5 7600 6-Core Processor CPU@3.80GHz, DDR5 4800MHz 16GBx2, OS-Ubuntu 22.04.3 LTS, Chipset-ASUS ProArt X670E-CREATOR WIFI STCOM.

7) Sufficient power is required for the stable operation of high-capacity SSDs, including the host device. To minimize system errors, please ensure the host device's power consumption requirements are met.

8) All documented endurance test results are in compliance with JESD218 Standards. Please visit www.jedec.org for detailed information on JESD218 Standards. TBW means Terabytes Written, Warranty provides coverage for the stated time period or the TBW, whichever comes first. Please refer to the detailed warranty statement here at <http://www.samsung.com/samsungssd>

PRODUCT LINEUP

Density	Model Name	Box Contents	Model Code
1TB (1,000GB*)	MZ-VAP1T0	Samsung SSD 9100 PRO 1TB Warranty Statement	MZ-VAP1T0BW MZ-VAP1T0B/AM
2TB (2,000GB*)	MZ-VAP2T0	Samsung SSD 9100 PRO 2TB Warranty Statement	MZ-VAP2T0BW MZ-VAP2T0B/AM
4TB (4,000GB*)	MZ-VAP4T0	Samsung SSD 9100 PRO 4TB Warranty Statement	MZ-VAP4T0BW MZ-VAP4T0B/AM
8TB (8,000GB*)	MZ-VAP8T0	Samsung SSD 9100 PRO 8TB Warranty Statement	MZ-VAP8T0BW MZ-VAP8T0B/AM

* GB: 1GB = 1,000,000,000 bytes. The actual usable capacity may be less than the labeled capacity.

For more information, including but not limited to the warranty provided for this product, and to download the latest software & manuals, please visit www.samsung.com/ssd and www.samsungssd.com.

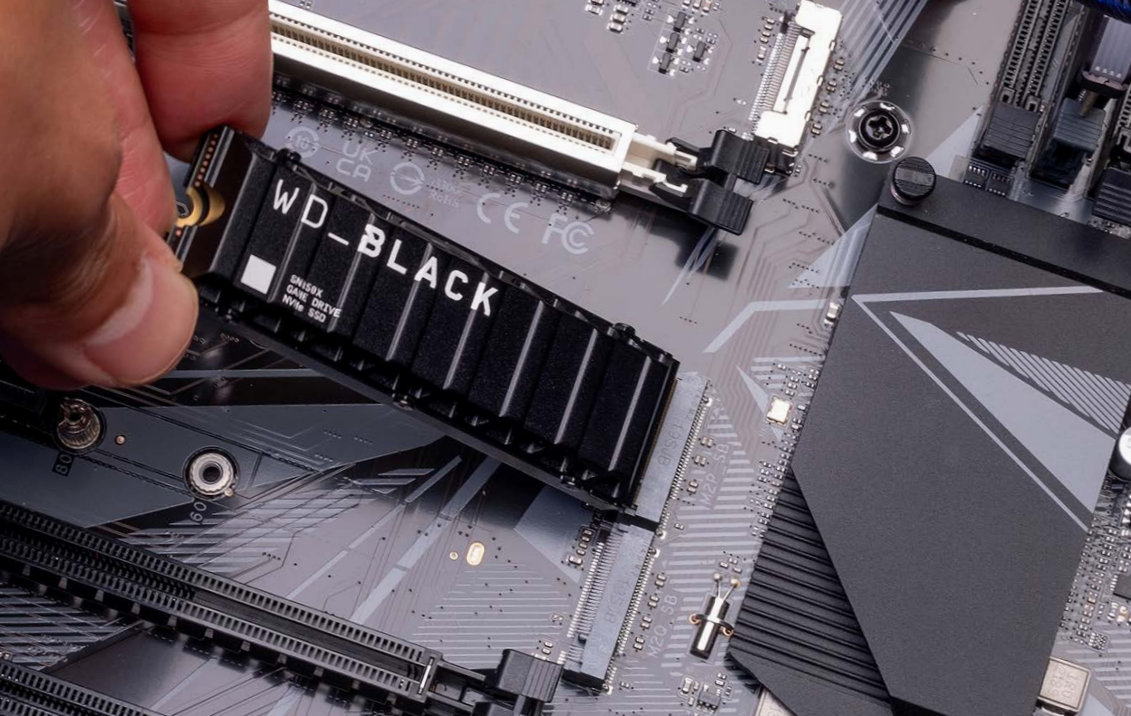
TEST CONFIGURATION

Below you will find a list of system configurations Samsung used to obtain the results reported in this Data Sheet. Performance/power data was measured with the SSD as a secondary drive in a fan cooling desktop system.

	Read/Write Performance	Power Consumption
Interface	PCIe 5.0 x4	PCIe 5.0 x4
OS	Linux Ubuntu 22.04.2	Linux Ubuntu 22.04.3 LTS
CPU	AMD Ryzen9 7950x 16-Core Processor CPU@4.5GHz	AMD Ryzen 5 7600 6-Core Processor CPU@3.80GHz
Memory	DDR5 4800MHz 16GBx2	DDR5 4800MHz 16GBx2
Chipset	ASRock-X670E-Taichi	ASUS ProArt X670E-CREATOR WIFI STCOM
Test Program	FIO 3.33	FIO 3.33

Revision History

Revision Number	Description	Revision Date
1.0	Initial Release	January, 2025
2.0	Update Information of the 8TB Capacity	September, 2025



WD_
BLACK™

SANDISK™

WD_BLACK™ SN850X NVMe™ SSD

M.2 2280 NVMe™ SSD GAME DRIVE

BUILT FOR ELITE GAMING.

Crush load times and slash throttling, lagging, and texture pop-ins with the WD_BLACK™ SN850X NVMe™ SSD. Available in capacities of up to 1TB through 8TB¹, this gaming drive is built for top-tier performance. With 7,300MB/s² speeds [1TB - 4TB models] and a heatsink option to maintain performance, you can enjoy your favorite games without worrying about lag. Plus, the downloadable WD_BLACK™ Dashboard [Windows® only] can automatically turn on Game Mode 2.0 features. The WD_BLACK™ SN850X NVMe™ SSD delivers a smooth, fast experience-and the innovations that let you compete at your absolute peak.

KEY FEATURES

- TRANSFORM YOUR PC. Get top-tier performance and ridiculously short load times for your gaming PC or workstation with insane speeds up to 7,300 MB/s² [1TB - 4TB models].
- MORE ROOM, MORE GAMES. Capacities up to 8TB¹ mean you get to download more games - and stop deleting your favorites.
- HEATSINK FOR THE WIN. All capacities have an optional heatsink version to help maintain peak performance through your most intense gaming sessions.
- DO MORE WITH DASHBOARD. The downloadable SANDISK® Dashboard [Windows® only] monitors your drive's health, lets you customize your RGB lighting [heatsink versions only], and can automatically detect games to turn on Game Mode 2.0.
- FUTURE-FORWARD FEATURES. The WD_BLACK™ SN850X SSD boasts a powerful suite of features, including Predictive Loading, Overhead Balancing, and Adaptive Thermal Management [ATM].³
- DIRECTSTORAGE SUPPORT. Improve load times and get precious time back with Microsoft® DirectStorage technology.

WD_
BLACK



PRODUCT FEATURES

GAIN THE ULTIMATE GAMING EDGE

Insane speeds up to 7,300MB/s² [1TB - 4TB models] deliver top-tier performance with ridiculously short load times - for the elite gaming experience you've been waiting for.

HEATSINK FOR THE WIN

All capacities have an optional heatsink version that's not only eye-catching, but also helps your rig maintain peak performance through your most intense gaming sessions.

MORE ROOM FOR MORE GAMES

Capacities up to 8TB¹ mean you get to keep more games at the ready - and get into the action even faster.

DO MORE WITH SANDISK® DASHBOARD

The downloadable SANDISK® Dashboard [Windows® only] monitors your drive's health, lets you customize your RGB lighting, and, exclusively on the SN850X SSD, enables Game Mode 2.0 to transform your gaming experience.

FUTURE-FORWARD FEATURES

The WD_BLACK™ SN850X SSD boasts a powerful suite of features, including Predictive Loading, Overhead Balancing, and Adaptive Thermal Management [ATM].³

DIRECTSTORAGE SUPPORTED

Improve load times and get precious time back with Microsoft® DirectStorage technology.

PRODUCT SPECIFICATIONS				
CAPACITIES ¹ :	8TB	4TB	2TB	1TB
MODELS WITH HEATSINK:	WDS800T2XHE	WDS400T2XHE	WDS200T2XHE	WDS100T2XHE
MODELS WITHOUT HEATSINK:	WDS800T2X0E	WDS400T2X0E	WDS200T2X0E	WDS100T2X0E
INTERFACE ⁴	PCIe® GEN4 16GT/S, UP TO 4 LANES			
NAND	SANDISK® TLC 3D NAND			
PERFORMANCE ²				
Sequential Read [up to]	7,200MB/s	7,300MB/s	7,300MB/s	7,300MB/s
Sequential Write [up to]	6,600MB/s	6,600MB/s	6,600MB/s	6,300MB/s
Random Read [up to]	1,200K IOPS	1,200K IOPS	1,200K IOPS	800K IOPS
Random Write [up to]	1,200K IOPS	1,100K IOPS	1,100K IOPS	1,100K IOPS
ENDURANCE ⁵ [TBW]	4,800	2,400	1,200	600
DIMENSIONS	Length:	Width:	Height:	Weight:
1 & 2TB MODELS WITHOUT HEATSINK:	80 ± 0.15mm	22 ± 0.15mm	2.38mm	7.5G ± 1g
4 & 8TB MODELS WITHOUT HEATSINK:	80 ± 0.15mm	22 ± 0.15mm	3.88mm	8.6G ± 1g
1 & 2TB MODELS WITH HEATSINK:	80 ± 0.20mm	23.40 ± 0.40mm	8.90 ± 0.40mm	24G ± 3g
4 & 8TB MODELS WITH HEATSINK:	80 ± 0.20mm	24.46 ± 0.48mm	10.31 ± 0.50mm	30.4G ± 3g
SECURITY SPECIFICATIONS	TCG OPAL V2.01			
SYSTEM COMPATIBILITY	Computer with M.2 [M-key] port, capable of taking M.2 2280 form factor Windows® 10+ PlayStation®5 [Heatsink model only]			
LIMITED WARRANTY ⁶	5 Years			
RoHS COMPLIANCE	Yes			
OPERATING TEMPERATURE ⁷	32°F TO 185°F [0°C TO 85°C]			
NON OPERATING TEMPERATURE ⁸	-40°F TO 185°F [-40°C TO 85°C]			

¹ 1TB = 1 trillion bytes. Actual user capacity may be less depending on operating environment.

² Based on read speed, unless otherwise stated. 1 MB/s = 1 million bytes per second. Based on internal testing; performance may vary depending upon host device, usage conditions, drive capacity and other factors.

³ Requires the downloadable WD_BLACK™ Dashboard [Windows® only].

⁴ PCIe® Gen4 storage technology requires a compatible motherboard. Backwards compatible with PCIe® Gen3 x4, PCIe® Gen3 x2, PCIe® Gen3 x1, PCIe® Gen2 x4, PCIe® Gen2 x2 and PCIe® Gen2 x1.

⁵ TBW [terabytes written] values calculated using JEDEC client workload [JESD219] and vary by product capacity.

⁶ 5 years or Max Endurance [TBW] limit, whichever occurs first. See support.wdc.com for region-specific warranty details.

⁷ Operational temperature is defined as temperature reported by the drive. Note that drive temperature readings are expected to be higher than ambient temperature when the SSD is placed inside a system. The SSD box package is rated up to 60°C.

⁸ Non-operational storage temperature does not guarantee data retention.

SANDISK™

July 2025

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